

Project Changes since Check 1

Since the first checkpoint, my primary shift has been moving from using Yelp mainly as a validation tool to using it as a complementary data source for analysis. I performed a 1-to-1 match between the Google Places (GP) and Yelp datasets, rather than treating them as separate sources, after identifying imbalances in the GP data (especially in price levels and service attributes). By merging the two platforms, I construct a more comprehensive feature set for each restaurant by combining price, ratings, and service offerings, which reduces the impact of missing values in either dataset.

Data Analysis and Visualizations Plan

My preliminary exploratory data analysis revealed that the Google Places dataset has an imbalanced price-level distribution, with “price level 4” notably underrepresented. To address this, I merged Google Places and Yelp records using restaurant names and geographic coordinates, producing a 1-to-1 matched sample of approximately 1,800 restaurants that provides a more reliable basis for modeling. I then created combined variables for visualization and analysis, including merged measures of price level, ratings, review volume, and service availability.

To transition from exploratory to polished visuals, I will create a set of maps and summary plots that communicate spatial concentration and SES gradients in restaurant characteristics. These include a scatter map of restaurant locations colored by price level to highlight clustering across Chicago, choropleth maps showing restaurant density, median price level, and mean rating by community area, and a heatmap comparing the share of restaurants offering delivery, takeout, dine-in, and reservations across neighborhood SES quartiles. Finally, I will run regression analyses to identify which restaurant characteristics are associated with higher ratings and to estimate how neighborhood SES covaries with restaurant ratings and price levels.